

CHAPTER - 4

IDENTIFYING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

The Alibaba B2B E-Commerce Database Management System, built around the JBL Bluetooth earbuds wholesale sourcing case study, is designed to efficiently manage suppliers, categories, products, customers, orders, order items, payments, and reviews. Functional dependencies help identify how non-key attributes relate to the primary key in each table, while redundancy analysis and data anomaly detection ensure that data remains consistent, accurate, and free of unnecessary duplication. A well-designed database of this kind improves query performance, maintains referential integrity, and supports reliable sourcing and sales analysis for wholesale buyers.

4.2 SUPPLIER TABLE

Functional Dependency

Supplier_ID → Supplier_Name, Location, MOQ_Policy, Certification_Status, Contact_No, Email

Entity Table – Supplier (Strong Entity)

Attribute	Key Type
Supplier_ID	Primary Key
Supplier_Name	Not Null
Location	Not Null
MOQ_Policy	-
Certification_Status	Check Key
Contact_No	Unique Key
Email	Unique Key

Redundancy Analysis

Supplier details are stored only once in the Supplier table. Products, Order_Items, and Reviews reference the supplier using Supplier_ID instead of repeating supplier information.

Data Anomaly

Anomaly Present: Yes — A Delete Anomaly risk exists — if a Supplier with existing Products or Reviews is deleted without RESTRICT/CASCADE rules in place, those child records become orphaned and lose their supplier reference.

4.2. CATEGORY TABLE

Functional Dependency

Category_ID → Category_Name, Description

Entity Table – Category (Strong Entity)

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Redundancy Analysis

Each category is stored only once. Products reference the category using Category_ID instead of repeating the category name and description.

Data Anomaly

Anomaly Present: No — Category is a fully independent strong entity. It can exist with or without Products, and deleting a Product never affects the Category record, so no insert, update, or delete anomaly occurs.

4.3. PRODUCT TABLE

Functional Dependency

Product_ID → Product_Name, Category_ID, Supplier_ID, Price, MOQ, Listing_Type, Key_Features, Units_Sold

Entity Table – Product (Strong Entity)

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Category_ID	Foreign Key
Supplier_ID	Foreign Key
Price	-
MOQ	Not Null
Listing_Type	Check Key
Key_Features	-
Units_Sold	-

Redundancy Analysis

Product details are stored only in the Product table. Order_Item records use Product_ID instead of repeating product name, price, or features for every order.

Data Anomaly

Anomaly Present: Yes — An Update Anomaly risk exists on Units_Sold — since this value duplicates a count derivable from Order_Item, it can go out of sync with actual sales unless updated automatically through a trigger.

4.4. CUSTOMER TABLE

Functional Dependency

Customer_ID → Business_Name, Contact_No, Email, Country, Address

Entity Table – Customer (Strong Entity)

Attribute	Key Type
Customer_ID	Primary Key
Business_Name	Not Null
Contact_No	Unique Key
Email	Unique Key

Country	-
Address	-

Redundancy Analysis

Customer/buyer information is stored only once in the Customer table. Orders and Reviews store only the Customer_ID, preventing duplicate customer records.

Data Anomaly

Anomaly Present: No — Customer is a fully independent, self-contained entity. Deleting an Order does not affect Customer data, and Customer attributes are not duplicated in any other table, so no anomaly occurs.

4.5. ORDER TABLE

Functional Dependency

Order_ID → Customer_ID, Order_Date, Order_Status, Total_Order_Value

Entity Table – Order (Strong Entity)

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	Check Key
Total_Order_Value	-

Redundancy Analysis

The Order table stores only order-level information. Customer details are referenced using the Customer_ID foreign key rather than being repeated.

Data Anomaly

Anomaly Present: Yes — An Update Anomaly risk exists — Total_Order_Value is a derived value duplicating the SUM of Quantity × Unit_Price from Order_Item, so it can become inconsistent if Order_Item changes are not cascaded to it.

4.6. ORDER_ITEM TABLE

Functional Dependency

Order_Item_ID → Order_ID, Product_ID, Supplier_ID, Quantity, Unit_Price

Entity Table – Order_Item (Weak Entity)

Attribute	Key Type
Order_Item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Supplier_ID	Foreign Key
Quantity	Check Key
Unit_Price	-

Redundancy Analysis

This table connects Orders, Products, and Suppliers, avoiding repeated product or supplier information inside the Order table itself.

Data Anomaly

Anomaly Present: No — All attributes (Quantity, Unit_Price) depend fully on the composite key (Order_ID, Product_ID, Supplier_ID). No redundant or derivable-only data is stored here, so no anomaly occurs.

4.7. PAYMENT TABLE

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount

Entity Table – Payment (Strong Entity)

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	Check Key
Payment_Date	-
Amount	-

Redundancy Analysis

Payment information is stored separately from the Order table, avoiding duplicate payment records for the same order (the One-to-One relationship ensures a single payment entry per order).

Data Anomaly

Anomaly Present: Yes — An Update Anomaly risk exists — Amount duplicates Order.Total_Order_Value, creating two sources of truth for the same monetary figure that must be kept in sync.

4.8. REVIEW TABLE

Functional Dependency

Review_ID → Customer_ID, Supplier_ID, Rating, Review_Text, Review_Date

Entity Table – Review (Strong Entity)

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Supplier_ID	Foreign Key
Rating	Check (1-5)
Review_Text	-
Review_Date	-

Redundancy Analysis

Review information is stored separately from Customer and Supplier tables, avoiding repetition of business or contact details for every review submitted.

Data Anomaly

Anomaly Present: No — Review_ID uniquely determines its own attributes with no dependency on other tables' data. Editing or deleting a review does not affect Customer or Supplier records, so no anomaly occurs.

4.9. CONCLUSION

The Alibaba B2B E-Commerce Database Management System is designed using proper functional dependencies and normalized tables for Suppliers, Categories, Products, Customers, Orders, Order_Items, Payments, and Reviews. Each entity stores only its relevant data, with foreign keys used to link related information instead of repeating it, thereby reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains referential integrity, and ensures an efficient and reliable database for managing wholesale sourcing, pricing comparisons, and order tracking on a B2B platform.